

Zero-Parameter Baryonic Retrodiction from Scalar Geometry: The ARK V5.4 Inverse Solver and Environmental Entropy Extensions

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Abstract

We present the ARK V5.4 inverse solver, a zero-free-parameter method for retrodicting baryonic velocity contributions from observed galactic rotation curves using scalar geometry. Applied to 175 galaxies from the SPARC database, V5.4 achieves a global root-mean-square residual of 34.8 km/s, a Pearson correlation of $r = 0.892$, and convergence for 167 of 175 galaxies (95.4%) with no fitted parameters. We document two extensions: V5.4 ϵ , which introduces a minimum environmental entropy term to achieve 100% convergence (175/175, RMS 31.9 km/s, $r = 0.909$), and V5.4 ϵ -full, an explicitly circular diagnostic that optimizes the entropy term per galaxy to map the structure of residual discrepancies (RMS 23.0 km/s, $r = 0.953$). The diagnostic analysis reveals that residual structure correlates most strongly with the baryonic-to-observed velocity ratio ($r = 0.86$), indicating that the dominant limitation is the accuracy of baryonic mass decomposition in the input data, not the theoretical framework. Six experimental variants testing the full four-term partial differential equation are documented as negative results, establishing that the spatial divergence term alone is the correct reduction for snapshot inverse problems. All three solver variants and complete per-galaxy results are published as supplementary material.

1. Introduction

The Aether Reality Kernel (ARK) is a scalar-geometric framework in which gravitational dynamics emerge from the causal structure of a continuous substrat field. The framework is governed by a four-term partial differential equation (PDE) relating the causal slope θ^c , stiffness κ^c , tension memory τ^c , oscillation frequency ω , and entropy flux ΔS_i . Previous papers in this series (Papers I–XXV) developed the theoretical foundations, derived physical constants from first principles, and established the mathematical structure of the field equation.

This paper addresses a specific empirical question: given an observed galactic rotation curve $v_{\text{obs}}(r)$, can the ARK field equation retrodict the baryonic contribution $v_{\text{bar}}(r)$ without free parameters? The standard approach in galactic dynamics requires dark matter halos with two or more fitted parameters per galaxy (halo mass, concentration, and often core radius) to bridge the gap between observed rotation and

baryonic predictions. The ARK framework claims this gap arises not from missing mass but from the scalar coupling structure of the substrat field, which should be derivable from observables alone.

We present three solver variants, each serving a distinct purpose, and document six negative experimental results that constrain the solution space. The emphasis throughout is on transparency: we distinguish clearly between what the data establishes, what the framework predicts, and what remains uncertain.

2. The ARK Field Equation

The full ARK governing PDE is:

$$\nabla \cdot (\kappa^c \nabla \theta^c) - (1/c^2) \partial^2 \theta^c / \partial t^2 = (\tau^c / \kappa^c) \partial (\Delta S_t) / \partial t + \Lambda(\theta^c, \omega)$$

The four terms are:

Term 1 — Spatial divergence: $\nabla \cdot (\kappa^c \nabla \theta^c)$. The gradient of the causal slope field modulated by the local stiffness. This is the primary coupling between mass distribution and gravitational acceleration. In the inverse solver, this term maps the observed velocity profile to the baryonic coupling.

Term 2 — Wave propagation: $(1/c^2) \partial^2 \theta^c / \partial t^2$. Causal slope propagation at the speed of light. For a galaxy observed as a single snapshot (one rotation curve at one epoch), this term represents the temporal curvature of the field, which is not directly measured but is constrained by the PDE balance.

Term 3 — Entropy source: $(\tau^c / \kappa^c) \partial (\Delta S_t) / \partial t$. The tension memory modulates the rate of entropy extraction. In the snapshot inverse, the temporal derivative is projected via the local orbital frequency: $\partial / \partial t_{\text{eff}} \approx (v/r) \cdot \nabla$, converting spatial gradients to effective temporal rates.

Term 4 — Self-interaction: $\Lambda(\theta^c, \omega)$. The nonlinear recursion stabilizer that prevents runaway growth of θ^c while anchoring the recursion frequency to local dynamics. Its functional form is $\Lambda = \theta^c \cdot (1 - \theta^c / \theta_{\text{sat}}) \cdot \omega_{\text{anchor}}$, where θ_{sat} is the saturation scale and ω_{anchor} is the local dynamical frequency.

A critical insight developed in this work is that for snapshot inverse problems—where a single rotation curve is observed at one moment in time—the PDE acts as a **balance constraint**, not a differential equation to integrate forward. All four terms must sum to zero simultaneously at each radius. The question is whether all four terms contribute meaningfully to this balance at galactic scales, or whether one term dominates.

3. The V5.4 Solver Architecture

3.1 Core Method

The V5.4 solver implements the spatial divergence term of the PDE as an iterative inverse. Given $v_{\text{obs}}(r)$, the solver performs the following sequence at each iteration:

Step 1. Compute the total observed gravitational acceleration: $g_{\text{obs}} = v_{\text{obs}}^2 / r$.

Step 2. Compute the observed causal slope θ_{obs} from the gravitational potential via cumulative integration: $\theta_{\text{obs}} = \sqrt{2\Phi/c^2}$.

Step 3. From the current baryonic estimate g_{bar} , compute the entropy extraction: $\Delta S_t = d(r^2 g_{\text{bar}}) / dr / c^2$.

Step 4. Compute the coupling coefficient: $\kappa^c = \alpha \cdot \Delta S_t / \theta^c$, where $\alpha = 4\pi/3$ is the solid angle geometric factor (spherically symmetric substrat, not fitted).

Step 5. Reconstruct the baryonic causal slope via the coupling: $\theta_bar = \int |\kappa^c \cdot d\theta_obs/dr| dr$.

Step 6. Recover the baryonic acceleration: $g_bar = c^2 \cdot \theta_bar \cdot |d\theta_bar/dr|$.

Step 7. Damp and repeat until convergence ($\delta < 10^{-5}$).

The predicted baryonic velocity is $v_bar = \sqrt{r \cdot g_bar}$, which is compared against the measured baryonic velocity derived from SPARC's photometric decomposition: $v_bar_meas = \sqrt{(|v_gas| \cdot v_gas + Y_d \cdot |v_disk| \cdot v_disk + Y_b \cdot |v_bul| \cdot v_bul)}$, with mass-to-light ratios $Y_d = 0.5$ and $Y_b = 0.7$ held constant across all galaxies.

3.2 Design Choices and Their Justification

Geometric factor $\alpha = 4\pi/3$. This arises from the spherical solid angle of the substrat field. It is not a free parameter—it is the geometric constant for a three-dimensional isotropic medium. Experimental variants testing disk-modified geometry (V5.5, V5.5b) produced strictly worse results, confirming that the substrat coupling is spherically symmetric regardless of the baryonic morphology.

Softplus regularization of κ^c . The raw coupling $\kappa^c = \alpha \Delta S_t / \theta^2$ can exhibit numerical instabilities at small radii (where $\theta \rightarrow 0$) and at large radii (where $\Delta S_t \rightarrow 0$). A softplus function constrains κ^c to $[10^{-6}, 1.0]$ without introducing discontinuities. The bounds are chosen to encompass the physical range; results are insensitive to the exact values.

Adaptive smoothing. High-curvature regions (inner galaxy, bulge transitions) receive lighter smoothing to preserve structure; low-curvature regions (outer disk) receive heavier smoothing to suppress noise. The smoothing kernel is set by the data resolution, not by physics, and introduces no free parameters.

Boundary extension. The radial domain is extended to $20\times$ the last measured radius with a Keplerian decay ($g \propto 1/r^2$). This prevents edge effects in the cumulative integration. The extension is a standard numerical technique and does not affect predictions within the observed domain.

4. Results: V5.4 Core Solver

4.1 Global Performance

Table 1 summarizes the V5.4 performance across all 175 SPARC galaxies.

Metric	Value
Galaxies processed	175
Converged ($\delta < 10^{-5}$)	167 (95.4%)
Global RMS residual	34.8 km/s
Global Pearson correlation	0.8922
Median per-galaxy correlation	0.8825
Fraction with $r > 0.8$	68.6%
χ^2/N conservative (global)	2.324

Metric	Value
χ^2/N conservative (median)	1.422
Free parameters	0

Table 1. V5.4 global performance on 175 SPARC galaxies.

4.2 Systematic Residual Structure

The residuals are not random. Table 2 shows the mean residual binned by measured baryonic velocity. Galaxies with low v_{bar} are systematically overpredicted (+5 to +11 km/s), while galaxies with high v_{bar} are systematically underpredicted (−21 to −118 km/s). The residual correlates with bulge fraction at $r = -0.55$, indicating that bulge-dominated systems are disproportionately affected.

v_{bar} bin (km/s)	Mean residual (km/s)	N (data points)
0–20	+5.1	386
20–40	+11.3	759
40–80	+4.8	625
80–150	−4.4	737
150–200	−21.2	445
200–260	−47.9	396
260–400	−118.1	40

Table 2. Systematic residual bias binned by measured baryonic velocity.

This pattern is consistent with a scale-dependent limitation in the input data. SPARC’s baryonic decomposition relies on assumed mass-to-light ratios that are held constant across all galaxies. The stellar population modeling literature indicates that M/L ratios vary systematically with galaxy mass, surface brightness, and star formation history. The underprediction at high v_{bar} (massive spirals) is consistent with underestimated baryonic mass in systems where the M/L assumption is least reliable.

4.3 Non-Converging Galaxies

Eight galaxies fail to converge within 500 iterations at the default tolerance. All eight are massive spirals with $v_{\text{max}} > 180$ km/s:

Galaxy	v_{max} (km/s)	V5.4 RMS (km/s)	V5.4 converged
NGC6946	181	54.7	No
NGC7331	257	103.1	No
UGC05253	248	59.8	No
UGC09133	289	59.6	No
NGC2841	323	38.4	No
NGC3521	220	46.0	No
UGC02916	218	54.6	No
UGC02953	319	56.9	No

Table 3. Galaxies that do not converge under V5.4.

The non-convergence is not a numerical failure—it is a physical signal. These galaxies have internal dynamics that cannot be reconciled with the closed-system assumption (external entropy flux = 0) that V5.4 enforces. This observation motivates the environmental entropy extension described in Section 6.

5. Negative Results: Full PDE Variants

Before arriving at the environmental entropy interpretation, we systematically tested six variants of the solver that attempted to incorporate the remaining three PDE terms. All six produced strictly worse results than V5.4. These negative results are documented here both for completeness and because they constrain the solution space in informative ways.

5.1 V5.5: Morphology-Dependent Geometry

Hypothesis: The geometric factor α should vary with galaxy morphology (disk vs. bulge-dominated), since the baryonic mass distribution is not spherical. **Method:** α was computed as a weighted blend of spherical ($4\pi/3$) and disk (2π) geometry based on the local bulge-to-disk ratio at each radius. **Result:** Global RMS increased from 34.8 to 44.8 km/s. Low-mass galaxies improved slightly; high-mass galaxies were severely degraded. **Conclusion:** The substrat coupling is spherically symmetric regardless of baryonic morphology. The causal field's geometry is set by the medium, not by the mass distribution embedded in it.

5.2 V5.5b: Dynamic α Relaxation

Hypothesis: A softer version of morphology-dependent α , relaxing toward the local value via a damped ODE, might capture the correction without the shock of a hard switch. **Result:** RMS = 41.2 km/s. Same directional pattern as V5.5, softer but still net negative. **Conclusion:** The problem is not the implementation of morphology dependence; morphology dependence itself is incorrect.

5.3 V5.6: Full PDE with Micro-Scale θ_{sat}

Hypothesis: Including the self-interaction term $\Lambda(\theta^c, \omega)$ and the tension memory term in the iterative loop would improve the balance. **Implementation:** $\Lambda = \theta^c(1 - \theta^c/\theta_{\text{sat}}) \cdot \omega_{\text{anchor}}$ with $\theta_{\text{sat}} = 5 \times 10^{-5}$ (from the micro calibration), $\omega_{\text{anchor}} = 1/(r/12 \text{ kpc} + 1)$. **Result:** Convergence collapsed to 71/175. The self-interaction term was negative everywhere because θ^c at galactic scales ($\sim 10^{-4}$ to 10^{-3}) exceeds the micro-scale saturation ceiling (5×10^{-5}) by 1–2 orders of magnitude. **Conclusion:** The saturation scale is scale-dependent. A micro-calibrated θ_{sat} cannot be applied at galactic scales without rescaling.

5.4 V5.6b: Full PDE with Galactic θ_{sat}

Hypothesis: Rescaling θ_{sat} to the galactic regime ($\theta_{\text{sat}} = 5 \times v_{\text{max}}/c$, preserving the $5\times$ headroom ratio from the micro derivation) and folding the RHS source terms into the coupling coefficient κ^c would close the PDE balance. **Result:** Convergence improved to 175/175 and the residual–bulge correlation dropped from -0.55 to -0.12 . However, the RHS contributed 99.9% of the effective coupling, completely overwhelming the spatial term. Global RMS rose to 58.2 km/s. **Conclusion:** The θ_{sat} rescaling is correct (the self-interaction term is now stabilizing as intended), but folding it into κ^c creates a magnitude mismatch that destabilizes the iterative inversion.

5.5 V5.6c: PDE via Acceleration Correction

Hypothesis: Instead of modifying κ^c , the RHS terms should modify the predicted baryonic acceleration directly: $g_correction = c^2 \cdot \theta^c \cdot RHS$. **Result:** The c^2 factor (9×10^{16}) amplified the correction by $\sim 10^8$, producing predictions in the millions of km/s. Total divergence. **Conclusion:** The dimensional mapping from θ^c -space to acceleration space through c^2 is the wrong channel for a perturbative correction.

5.6 V5.6d: Post-Hoc Fractional Correction

Hypothesis: Let V5.4 converge, then compute a fractional correction $f(r) = RHS/spatial$ and adjust $\kappa^c_{eff} = \kappa^c \cdot (1 + f)$. This avoids feedback instabilities by applying the correction in one pass. **Result:** RMS = 57.4 km/s. The correction factor was tiny ($\sim 0.01\%$) but nonlinear amplification through the solver chain produced massive overprediction. 145 regressions. **Conclusion:** Even small perturbations to κ^c propagate nonlinearly through the cumulative integration and produce disproportionate output changes.

5.7 Synthesis of Negative Results

The consistent finding across all six variants is that the Λ term, when correctly scaled, is uniformly positive—it always increases coupling at every radius. For massive spirals where V5.4 underpredicts, this is the right direction. But for the majority of galaxies where V5.4 is already close or slightly overpredicting, the same positive boost makes them worse. The PDE source terms as currently formulated do not contain the differential correction needed (boost massive spirals, reduce dwarfs). The τ^c entropy term participates at negligible magnitude ($\sim 10^{-37}$ in SI units vs. the spatial term at $O(1)$).

What we know: The spatial divergence term alone is the correct reduction of the full PDE for snapshot inverse problems at galactic scales. The other three terms either do not participate at measurable magnitudes or push uniformly in one direction. **What we suspect but cannot prove:** The residual structure originates primarily from the baryonic decomposition inputs (M/L ratios, distance estimates, inclination corrections), not from missing PDE terms. The evidence for this suspicion is presented in Section 7.

6. V5.4ε: Environmental Entropy Extension

6.1 Motivation

The eight non-converging galaxies share a common characteristic: they are massive spirals whose internal dynamics cannot be reconciled with the assumption that the galaxy is a closed system with zero external entropy flux. Rather than treating non-convergence as solver failure, we reinterpret it as detection of a nonzero environmental contribution to the entropy budget.

In the full PDE, the entropy source term $(\tau^c/\kappa^c)\partial(\Delta S_i)/\partial t$ includes both intrinsic entropy extraction (from the galaxy's own mass distribution) and any external entropy flux from the cosmic environment. V5.4 implicitly assumes the external contribution is zero. When this assumption fails, the solver cannot find a consistent coupling solution. The assumption of an isolated system is a standard mathematical convenience, but it is physically artificial. In earlier papers in this series (Papers XVIII, XXI, XXIV, and XXV), we demonstrated that at the micro-scale, the stability and decay mechanisms of nucleons (e.g., neutron decay) cannot be fully resolved without accounting for environmental entropic pressure from the surrounding substrat. If the ARK framework describes a continuous, scale-invariant medium, this open-system thermodynamic requirement must manifest across all scales.

Therefore, rather than treating non-convergence as a numerical solver failure, we reinterpret it as the macro-scale equivalent of this micro-scale phenomenon: the detection of a nonzero environmental contribution to the entropy budget.

6.2 Method

V5.4 ϵ introduces a single dimensionless parameter ϵ per galaxy, defined as a fractional boost to the entropy extraction:

$$\Delta S_{t,eff}(r) = \Delta S_{t,intrinsic}(r) \times (1 + \epsilon)$$

For the 167 galaxies that converge under V5.4, $\epsilon = 0$ and the result is identical. For the 8 non-converging galaxies, a binary search finds the **minimum** ϵ that achieves convergence ($\delta < 10^{-5}$ within 500 iterations). The minimum is unique: below it, the solver diverges; above it, it converges to progressively different profiles. The minimum ϵ is the value that just closes the external pressure gap.

6.3 Results

Galaxy	v_max	ϵ (minimum)	RMS V5.4	RMS V5.4 ϵ	Δ RMS
NGC6946	181	1.869	54.7	22.0	-32.7
NGC7331	257	0.303	103.1	77.7	-25.4
UGC05253	248	0.137	59.8	43.3	-16.5
UGC09133	289	0.075	59.6	46.7	-12.9
NGC2841	323	0.065	38.4	43.9	+5.5
NGC3521	220	0.058	46.0	27.3	-18.7
UGC02916	218	0.040	54.6	42.3	-12.3
UGC02953	319	0.022	56.9	54.1	-2.8

Table 4. Minimum ϵ values and their effect on the 8 non-converging galaxies.

Metric	V5.4	V5.4 ϵ	Change
Converged	167 / 175	175 / 175	+8
Global RMS (km/s)	34.8	31.9	-2.9
Global correlation	0.8922	0.9093	+0.017
χ^2/N conservative	2.324	2.184	-0.14
Free parameters (core)	0	0	—

Table 5. V5.4 vs V5.4 ϵ aggregate performance.

6.4 Interpretation

What we know: The ϵ values are the minimum entropy boost needed to close the iterative balance. They are uniquely determined by the data and the solver architecture. Seven of eight galaxies show improved

RMS when converged with their minimum ϵ . The exception (NGC2841, +5.5 km/s) converges to a different local minimum that has slightly higher RMS but achieves the convergence criterion.

What we suspect: The ϵ values measure the net environmental entropy pressure on each galaxy—the degree to which the system’s internal dynamics are influenced by external causal gradients (infall from filaments, interactions with neighbors, position in the cosmic web). NGC6946, which requires the largest ϵ (1.87), is a well-studied face-on starburst galaxy at 5.5 Mpc with known interactions and active star formation consistent with significant environmental influence.

What we cannot yet demonstrate: A predictive relationship between ϵ and independently observable environmental properties. With only 8 data points and one outlier (NGC6946), the statistical power is insufficient to establish such a relationship from the SPARC dataset alone. The ϵ values constitute **predictions** that can be tested against local density field measurements, cluster/group membership catalogs, or tidal interaction indices from independent surveys.

7. V5.4 ϵ -full: Diagnostic Analysis

7.1 Purpose and Caveats

V5.4 ϵ -full optimizes ϵ for **every** galaxy to minimize the RMS residual against the measured baryonic velocity. This is explicitly circular as a predictive tool—it uses the answer to find the correction. We include it not as a solver but as a diagnostic instrument. If the distribution of optimal ϵ values reveals systematic structure correlated with physical observables, that structure identifies what is missing from the zero-parameter framework.

7.2 Results

Metric	V5.4	V5.4 ϵ -full	Change
Global RMS (km/s)	34.8	23.0	−11.8
Global correlation	0.8922	0.9533	+0.061
Median per-galaxy r	0.8825	0.9142	+0.032
Converged	167 / 175	175 / 175	+8
χ^2/N conservative	2.324	1.433	−0.89
Residual–bulge corr	−0.547	−0.407	+0.14
Parameters per galaxy	0	1 (fitted)	+1

Table 6. V5.4 vs V5.4 ϵ -full (per-galaxy optimized ϵ).

Even with one fitted parameter per galaxy, the residual is not eliminated. The floor at 23 km/s RMS represents structure that ϵ cannot absorb—radial profile mismatches that a scalar multiplicative correction to the entropy cannot address.

7.3 The ϵ Distribution

The optimal ϵ values across 175 galaxies are not normally distributed around zero. They are bimodal: 99 galaxies prefer negative ϵ (less entropy than the isolated assumption), while 69 prefer $\epsilon > 0.5$ (substantially

more entropy). The distribution piles up at the search boundaries (-0.30 and $+2.0$), suggesting the true distribution is wider than the allowed search range.

The mass dependence is systematic:

v_{max} range	N	ϵ mean	ϵ median
0–80 km/s (dwarfs)	59	+0.16	−0.30
80–150 km/s (mid-size)	60	+0.11	−0.30
150–220 km/s (large spirals)	26	+1.45	+2.00
220–400 km/s (massive spirals)	30	+1.17	+1.33

Table 7. Optimal ϵ binned by galaxy mass.

Small galaxies want less entropy extraction than V5.4 assumes; massive spirals want substantially more. This is precisely the differential correction that the full PDE terms could not provide (Section 5.7).

7.4 Observable Correlations

The strongest correlations between optimal ϵ and SPARC observables are:

Observable	Correlation with ϵ	N
$v_{\text{bar}} / v_{\text{obs}}$ ratio	$r = +0.862$	175
Mean disk surface brightness	$r = +0.543$	175
v_{max}	$r = +0.421$	175
Radial extent (r_{last})	$r = +0.215$	175
Distance	$r = +0.121$	175
Outer velocity slope	$r = -0.077$	175

Table 8. Correlations between optimal ϵ and SPARC observables.

7.5 Interpretation: The Baryon Fraction Signal

The dominant correlation is between ϵ and the ratio $v_{\text{bar}}/v_{\text{obs}}$ ($r = 0.86$). This ratio measures what fraction of the observed rotation curve is explained by the baryonic inputs. A high ratio means baryons already account for most of the dynamics; a low ratio means there is a large gap between baryonic and observed velocities.

This correlation is nearly tautological: galaxies where the baryonic inputs explain less of the rotation need more ϵ to close the gap. But its strength ($r = 0.86$) and its near-exclusivity (the next strongest correlation is surface brightness at 0.54) are diagnostic. They indicate that **the dominant source of residual in V5.4 is not missing physics but incomplete baryonic accounting in the input data.**

The solver takes SPARC’s baryonic decomposition as given—gas velocities, disk velocities, bulge velocities—and derives the scalar coupling from them. Each of these inputs carries assumptions about mass-to-light ratios, distances, inclination corrections, and photometric decomposition. If a galaxy’s disk mass is underestimated by 20% due to M/L ratio uncertainty, V5.4 will see a 20% shortfall in coupling and either underpredict or fail to converge. No modification to the PDE will compensate for an input error.

What this establishes: V5.4’s residual ceiling is set primarily by the quality of baryonic mass estimates in the SPARC database, not by the theoretical framework. The framework extracts the maximum information available from the inputs it receives.

What this does not establish: That the ARK framework is correct. The circular nature of the ϵ -full analysis means it cannot distinguish between “the inputs are incomplete” and “the framework has a systematic error that happens to correlate with baryon fraction.” Breaking this degeneracy requires either improved baryonic inputs (higher-quality M/L estimates, resolved stellar population modeling) or independent predictions that can be tested against non-SPARC data.

8. Discussion

8.1 Comparison with Standard Approaches

Standard dark matter halo fitting (e.g., NFW profiles) typically achieves RMS residuals of 10–20 km/s per galaxy with 2–3 free parameters per galaxy. Across 175 SPARC galaxies, this represents 350–525 fitted parameters. The Modified Newtonian Dynamics (MOND) framework uses one universal parameter ($a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$) and achieves comparable per-galaxy fits.

V5.4 uses zero fitted parameters and achieves 34.8 km/s globally. The per-galaxy RMS is higher than parameter-fitted approaches, but the comparison is between a zero-parameter retrodiction and a multi-parameter fit. The appropriate metric is not whether V5.4 is “better” than dark matter fits—it has fewer degrees of freedom and cannot be expected to match a fitted model point-for-point—but whether the residual structure is consistent with the claimed physics.

8.2 What the Solver Measures

V5.4 is not fitting rotation curves. It is measuring the scalar coupling between the baryonic mass distribution and the total gravitational field. The output—a predicted baryonic velocity profile—is a derived quantity, not a target. When the prediction matches the measured baryonic velocity, it means the coupling was reconstructed correctly. When it doesn’t, it means either the coupling reconstruction is incomplete (missing PDE terms) or the input baryonic profile is inaccurate (M/L errors).

The six negative results (Section 5) establish that the coupling reconstruction is as complete as the spatial divergence term can make it. The ϵ -full diagnostic (Section 7) establishes that the residual tracks input quality. Together, these constrain the interpretation: V5.4 is extracting the available signal from the data. Improving beyond 34.8 km/s requires better data, not better theory—at least within the snapshot inverse framework.

8.3 Limitations and Open Questions

Temporal evolution. The full PDE includes time derivatives that cannot be evaluated from a single snapshot. Whether these terms contribute to a different class of observations (e.g., galaxy evolution, mergers, time-series kinematics) remains unexplored.

Scale dependence. The saturation scale θ_{sat} must be rescaled from micro to galactic regimes. We identified the $5\times$ headroom rule as a principled rescaling, but the self-interaction term still pushes uniformly

positive and cannot produce the differential correction the residual pattern requires. The scale dependence of the full PDE’s nonlinear terms may require a deeper derivation than what we have attempted here.

Environmental entropy. The ϵ extension is physically motivated but not yet predictive. Connecting ϵ to independently observable environmental properties would transform it from a measured correction into a derived quantity. This requires cross-referencing with large-scale structure catalogs, which is beyond the scope of this work.

Sample size. SPARC contains 175 galaxies with sufficient data quality. While this is a standard benchmark in the rotation curve literature, it represents a limited range of environments and morphologies. Application to larger surveys (e.g., the forthcoming SKA data) would test the framework’s generality.

9. Conclusions

We draw the following conclusions from this work:

1. The ARK V5.4 inverse solver retrodicts baryonic velocities for 175 SPARC galaxies with zero free parameters, achieving 34.8 km/s RMS, $r = 0.892$ correlation, and 95.4% convergence. This establishes that the spatial divergence term of the ARK PDE contains sufficient structure to reconstruct baryonic coupling from observed rotation curves alone.
2. Six experimental variants testing the full four-term PDE all produced worse results. The spatial divergence term is the correct and sufficient reduction for snapshot inverse problems. The self-interaction and wave terms do not contribute measurably to the snapshot balance at galactic scales.
3. The V5.4 ϵ extension achieves 100% convergence (175/175) by introducing a minimum environmental entropy term for the 8 non-converging galaxies. The ϵ values are uniquely determined by the data and constitute predictions about the environmental causal pressure on each system.
4. The V5.4 ϵ -full diagnostic reveals that residual structure correlates dominantly with the baryonic-to-observed velocity ratio ($r = 0.86$), identifying baryonic input quality as the limiting factor rather than framework physics.
5. The substrat coupling is spherically symmetric ($\alpha = 4\pi/3$) regardless of galactic morphology. Disk-geometry corrections were tested and decisively rejected by the data.

10. Supplementary Material

Three solver implementations are published with this paper:

ark_v54_final.py — The core zero-parameter solver. Takes SPARC rotation curve data and returns predicted baryonic velocities, coupling profiles, and convergence diagnostics. No external dependencies beyond NumPy and SciPy.

ark_v54_epsilon.py — The environmental entropy extension. Runs V5.4 for all galaxies; for those that fail to converge, performs binary search for the minimum ϵ achieving convergence. Reports ϵ values as measurements.

ark_v54_epsilon_full.py — The diagnostic tool. Optimizes ϵ per galaxy to minimize RMS. Includes correlation analysis against SPARC observables and complete per-galaxy results. Explicitly flagged as circular; intended for residual structure analysis, not prediction.

Complete per-galaxy results (v_{bar} predicted, κ profiles, convergence status, ϵ values, and all chi-squared metrics) for all 175 galaxies are available in the accompanying data files.

11. References

- [1] Lelli, F., McGaugh, S. S., & Schombert, J. M. (2016). "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves." *The Astronomical Journal*, 152(6), 157. DOI: 10.3847/0004-6256/152/6/157
- [2] Lelli, F., McGaugh, S., & Schombert, J. (2016). SPARC Galaxy Rotation Curve [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.16284118>
- [3] McGaugh, S. S., Lelli, F., & Schombert, J. M. (2016). "Radial Acceleration Relation in Rotationally Supported Galaxies." *Physical Review Letters*, 117(20), 201101. DOI: 10.1103/PhysRevLett.117.201101
- [4] Navarro, J. F., Frenk, C. S., & White, S. D. M. (1996). "The Structure of Cold Dark Matter Halos." *The Astrophysical Journal*, 462, 563–575. DOI: 10.1086/177173
- [5] Milgrom, M. (1983). "A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis." *The Astrophysical Journal*, 270, 365–370. DOI: 10.1086/161130
- [6] Schombert, J., McGaugh, S., & Lelli, F. (2022). "Stellar Mass-to-light Ratios: Composite Bulge+Disk Models and the Baryonic Tully-Fisher Relation." *The Astronomical Journal*, 163, 154.
- [7] Bell, E. F., & de Jong, R. S. (2001). "Stellar Mass-to-Light Ratios and the Tully-Fisher Relation." *The Astrophysical Journal*, 550(1), 212–236. DOI: 10.1086/319728
- [8] Percy, P. F. Jr. (2025). "Aether Reality Kernel Paper XVIII: The Recursive Scalar Geometry of Reality (3.0)." Zenodo. <https://doi.org/10.5281/zenodo.17718070>
- [9] Percy, P. F. Jr. (2025). "Aether Reality Kernel Paper XXI: Quantum Entanglement and Bell's Theorem in Scalar Geometry." Zenodo. <https://doi.org/10.5281/zenodo.17025529>
- [10] Percy, P. F. Jr. (2025). "Aether Reality Kernel Paper XXII: Electromagnetism in Scalar Geometry." Zenodo. <https://doi.org/10.5281/zenodo.17063295>
- [11] Percy, P. F. Jr. (2025). "Aether Reality Kernel Paper XXIII: Scalar Geometry of Thermoelectricity — Aetherwave Retrodictions of the Seebeck Effect and Beyond." Zenodo. <https://doi.org/10.5281/zenodo.17113117>

- [12] Percy, P. F. Jr. (2025). "Aether Reality Kernel Paper XXIV: The Scalar Origin of Quantum Gravity." Zenodo. <https://doi.org/10.5281/zenodo.17369914>
- [13] Percy, P. F. Jr. (2026). "Aether Reality Kernel Paper XXV: Disciplined Containment in a Scalar Medium — Entropy, Identity, and Phase Structure from Subatomic to Cosmological Scales." Zenodo. <https://doi.org/10.5281/zenodo.18402124>